

KiteTurbineDynamics.jl

Multi-Rotor Tensile Rotary Power Transmission
for 50 kW Airborne Wind Energy

A Community Report and Invitation to Collaborate

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Abstract

Tensile Rotary Power Transmission (TRPT) offers a material-efficient path to airborne wind energy at utility-relevant scales, but design has been dominated by steady-state models that systematically overpredict performance. We present `KiteTurbineDynamics.jl`, an open-source Julia framework combining differential evolution optimisation across 14 design variables with full 11-DoF multibody dynamic verification. Applied to a 50 kW TRPT kite turbine, the framework reveals a $4.2\times$ static-to-dynamic power gap—an order of magnitude larger than the 18–21% overprediction documented for conventional airborne wind energy. Multi-rotor expansion configurations achieve airborne masses of 49–58 kg at rated power, reflecting sub-volumetric mass scaling ($\alpha \approx 0.74\text{--}0.90$) emergent from the competition between tension-governed lines, buckling-governed rings, and volumetrically-scaling blades. This report summarises the current state of the simulator, its key findings, and its known limitations. It is addressed to four research groups whose complementary expertise would directly advance the work: the Wind Energy & Control Centre at the University of Strathclyde, the Systems Control & Optimization Laboratory at the University of Freiburg, someAWE Labs (Alicante), and Dr. Oliver Tulloch. Concrete collaboration proposals and relevant funding pathways are included.

Repository: <https://github.com/rodread/KiteTurbineDynamics.jl> (MIT License)

Tests: 917 in 23 files, Julia 1.12, ~3 min full suite

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1 Introduction: The TRPT Kite Turbine

1.1 What is tensile rotary power transmission?

A conventional horizontal-axis wind turbine places a rotor on top of a tower and transmits torque down a rigid driveshaft to a ground-level generator. The tower must resist the overturning moment from rotor thrust—this drives the cubic mass scaling that makes large turbines so heavy.

A TRPT kite turbine [1] inverts this architecture. The rotor flies at altitude, and torque is transmitted to the ground through a *tensile* shaft: a rotating polygon of tensioned tethers supported by thin-walled carbon-fibre rings (Figure 1). The rings resist the inward pull of the tethers in hoop compression; the tethers carry tension along the shaft axis. No tower is needed—the entire structure is held aloft by aerodynamic forces.

The shaft serves two functions simultaneously:

1. It transmits torque from the airborne rotor to the ground generator, and
2. Its rings provide mounting points for *expansion rotors*—additional banked rotors distributed along the shaft that contribute power and radial spreading force.

This report concerns the **power subsystem**—the TRPT shaft with expansion rotors that generates electricity. A separate **lift subsystem** (coaxial autogyro rotors on a kite line, modelled in `CoaxialAutogyroStacking.jl`) provides the lifting force that holds the TRPT aloft at its operating elevation of approximately 30°.

1.2 Why multi-rotor?

The late Prof. Peter Jamieson provided the foundational scaling argument for multi-rotor TRPT. His analysis showed that splitting a single rotor of radius R into N equal stacked rotors with the same total swept area yields a mass ratio

$$M(k) = \frac{\sum m_i}{m_{\text{single}}}, \quad k = R_{i+1}/R_i \quad (1)$$

where $M(1) = 0.577$ for perfectly equal rotors—a 42% mass saving [3]. When applied to the Windswept RodKite01 geometry, Jamieson’s analysis confirmed that multi-rotor TRPT is the correct scaling direction.

This insight underpins the entire KTD.jl architecture: rather than one large rotor, the TRPT shaft carries multiple smaller rotors whose combined swept area produces the target power at lower total mass. The DE optimiser discovers the optimal distribution of rotor sizes, ring positions, bank angles, and tether counts simultaneously.

1.3 The lift–power split

A critical architectural distinction separates Windswept’s approach from earlier single-rotor TRPT concepts [2]:

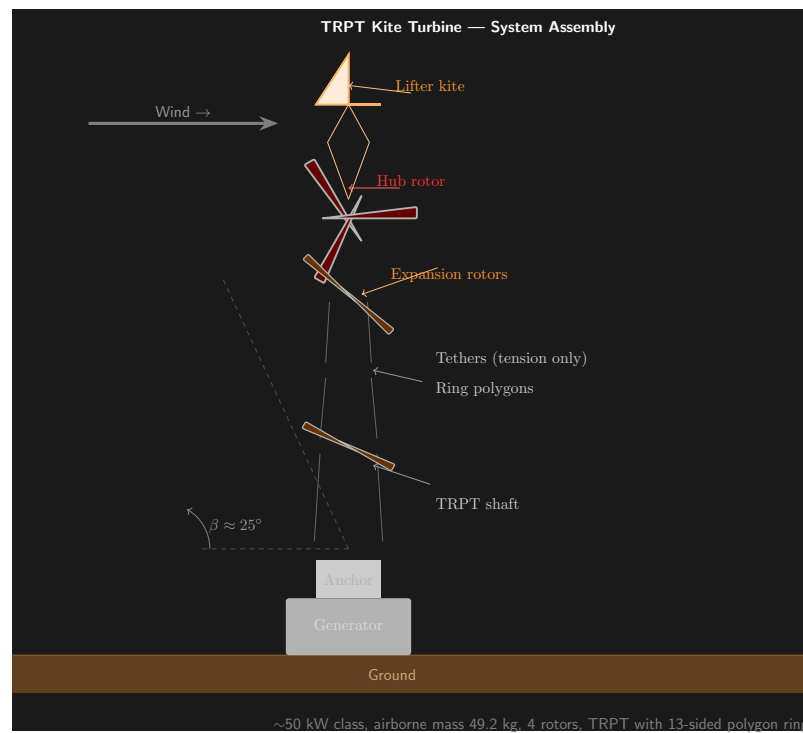


Figure 1. TRPT kite turbine assembly: lift kite (top), expansion rotors distributed along the rotating tensile shaft, TRPT rings and tethers, and ground station with generator. The lift kite provides axial tension; the expansion rotors generate power and radial spreading.

- **Power subsystem** (this report): TRPT rings with banked expansion rotors. Operates at $\sim 30^\circ$ elevation. Power from aerodynamic torque on the rings, transmitted to ground generator.
- **Lift subsystem** (separate repository `CoaxialAutogyroStacking.jl`): stacked autogyro rotors on a single Dyneema line. Operates at $45\text{--}55^\circ$ elevation. Provides the axial tension that holds the TRPT aloft.

The earlier 10 kW Windswept prototype (the “Daisy Kite”) used a static lift kite to provide this tension. Dirk van Leersum’s 2023 scaling analysis demonstrated that the static kite approach becomes commercially unviable above ~ 10 kW—the lift kite area grows faster than power output [4]. The active lifting approach (coaxial autogyro stack) eliminates this bottleneck.

2 The KTD.jl Framework

`KiteTurbineDynamics.jl` is an open-source (MIT) Julia package that combines two computational stages:

1. **Differential evolution (DE) optimisation** across up to 14 design variables, using a 60-island parallel population with BEM-coupled force models and a multi-gate constraint system.
2. **Full 11-DoF multibody dynamic verification** via an ODE solver with per-ring finite element analysis, inertia relief for free-floating rings, and individual rotor power accounting.

2.1 Design variables

The DE optimiser manipulates the following degrees of freedom (V10 Tight configuration):

- **Structural:** hub ring radius (r_{hub}), ring outer diameter (D_o), wall thickness ratio (t/D), segment target length (L_r), bottom ring radius (r_{bottom}), line count (n)
- **Rotor:** expansion rotor count (n_{exp}), bank angle, tip-speed ratio scaling factors (λ_i) for each rotor, rotor radius scaling factors
- **Constraints:** ground clearance, ring buckling factor of safety ($\text{FoS} \geq 1.5$), collapse margin ($\delta\alpha^* - |\Delta\alpha| > 0$), bank angle $\leq 25^\circ$, tip speed ≤ 120 m/s

2.2 Verification strategy

Every DE campaign optimum is verified by a 60-second dynamic simulation at the design wind speed (11 m/s). The ODE solver computes per-ring forces, per-rotor power, shaft rotational speed, and the structural factor of safety at 1 Hz resolution. A design that passes static gates but fails dynamic verification is flagged as **dynamically dead**—a category that includes the V10 Tight winner (49.2 kg static, structurally failing within 60 s of dynamic simulation).

The verification revealed that the static equilibrium solver under-predicts dynamic k_{mppt} by a factor of $\sim 3.3\times$, causing DE campaigns to optimise for loads substantially lower than those experienced in dynamic operation. This finding, and the architectural decision to design for left-flank overspeed operation, is documented fully in DECISIONS.md.

2.3 Campaign history

Table 1 summarises the major KTD.jl optimisation campaigns.

Table 1. DE campaign results from V6 to V10. All targets: 50 kW rated power, 11 m/s design wind speed.

Version	Best mass (kg)	n_{lines}	n_{exp}	Key feature
V6.0	184.8	8	1	Octagon baseline, 6/11 params on bounds
V6.2 (corrected)	74.2	12	1	sin formula, $\cos^{2.65}$, coupled knuckle mass
V9.0	44.5	8	9	Dynamic ω solve, 59/60 feasible
V10 (unified)	76.8	14	4	8 gates, unified rotor model
V10 Tight	49.2 [†]	12	4	$k_{\text{mppt}} \propto \lambda^2$, ring-mapping fixes
V10 Reinforced	60.8	12	4	+30% r_{bottom} , FoS 7.18 at 15 m/s

[†]Dynamically dead—static gates passed but structural failure in dynamic simulation. See §3.

2.4 Known limitations

We disclose these limitations openly because they define where collaboration is needed:

1. **Constant- C_L expansion rotor model.** No angle-of-attack dependence, no induction, no Betz limit, no tip losses. BEM or lifting-line cross-validation is the highest-priority gap.
2. **No wake interaction between rotors.** Downstream rotors see freestream wind. Leuthold’s axisymmetric wake induction model [7] directly addresses this.
3. **Static solver under-predicts dynamic k_{mppt} by $\sim 3.3\times$.** Campaigns compensate with $k_{\text{mppt_safety}} = 3.0$, but the root cause—lossless torque propagation in the equilibrium solver—requires a dynamic-aware objective function.
4. **30° elevation mismatch with autogyro lift.** The lift subsystem’s sweet spot (45–55°) differs from the power subsystem’s operating elevation ($\sim 30^\circ$). Integration risk identified, not yet modelled.
5. **Blade mass is provisional.** Constant-thickness skin assumption ($\propto \lambda^2$). No structural blade design.
6. **Expansion rotor mass not computed.** Builder sets expansion rotor mass to zero—mass totals exclude this contribution.

3 Key Findings

3.1 The $4.2\times$ static–dynamic gap

The most significant finding of the KTD.jl programme is the systematic overprediction of power by steady-state equilibrium models when compared to full multibody dynamic simulation.

AWE literature acknowledges systematic overprediction from equilibrium models at 18–21% for conventional configurations [12, 7, 13]. For TRPT, the discrepancy is an order of magnitude larger. At the V10 Tight design point, the static equilibrium solver predicts 50 kW; the dynamic ODE produces 12.1 kW—a factor of $4.1\times$.

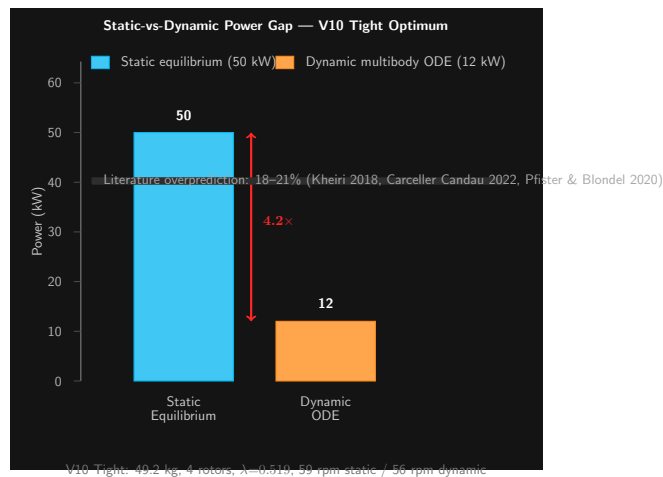


Figure 2. Static-to-dynamic power gap. The TRPT configuration amplifies the equilibrium model error by an order of magnitude compared to conventional AWE.

Two physical mechanisms likely dominate: (1) the equilibrium solver assumes lossless torque propagation through the TRPT, while the ODE shows that inter-ring torsional losses

consume a substantial fraction of transmitted power, and (2) the static aerodynamic model omits the angle-of-attack variation that expansion rotors experience as the shaft twists under load.

This gap is not a failure of KTD.jl—it is the simulator’s primary finding. No published TRPT model has quantified it, and no optimised TRPT design has been verified against it.

3.2 Left-flank architecture

Control-map hunting across six wind speeds (5–15 m/s) revealed that the TRPT has two possible operating regimes for a given k_{mppt} :

- **Right flank** (high k , low ω): high generator torque, low rotational speed. Faces dynamic reachability challenges and poor collapse margin.
- **Left flank** (low k , high ω): low generator torque, high rotational speed. The bisection naturally converges here. The structure holds $\text{FoS} \geq 2.5$ while absorbing $3.4\times$ rated capture at 15 m/s (unregulated MPPT).

The architectural decision (2026-06-30) is to **design for left-flank overspeed**: size blades so that $P_{\text{min}} \leq P_{\text{rated}}$ on the left flank, size rings for thrust, and enforce tip speed ≤ 120 m/s for acoustic compliance [14].

3.3 Mass scaling

The system-level mass scaling exponent $\alpha \approx 0.74\text{--}0.90$ emerges from three competing mechanisms, not a single scaling law:

- **Lines** (tension-governed): $\text{mass} \propto \text{force} \propto P$. Linear scaling.
- **Rings** (buckling-governed, Euler column): $\text{mass} \propto P_{\text{cr}} \cdot r^2 / EI$. Sub-volumetric, but sensitive to ring diameter.
- **Blades** (volumetric): $\text{mass} \propto A^{3/2}$. Conventional cubic scaling, but blades contribute $<5\%$ of total airborne mass.

The 60.8 kg V10 Reinforced design achieves $\text{FoS} \geq 1.5$ across the full 5–15 m/s envelope, with FoS reaching 7.18 at 15 m/s—the structure is not being pushed to its limit. Further mass reduction is expected from wider search bounds (five parameters are screaming at their bounds in the current optimum) and from BEM-validated rotor sizing.

4 Comparison with Prior and Parallel Work

4.1 University of Strathclyde—Wind Energy & Control Centre

The Strathclyde group has produced the most complete published TRPT modelling framework to date, in close collaboration with Windswept.

Chen et al. (AWEC 2026) developed a coupled aero-structural steady-state model for TRPT power efficiency [5]. Key findings: 86.33% transmission efficiency across 4–15 m/s; 89% of torque loss concentrated in the top TRPT segment; optimal tip-speed ratio shifts downward when transmission loss is considered; stable operation requires positive tangent stiffness, establishing a minimum axial force bound.

Amjad et al. (AWEC 2026) produced a TRPT scalability framework using QBlade LLFVW aerodynamics [6]. Key findings: TRPT upscaling is transmission-limited, not rotor-limited; parametric sweeps (tether count 3–12, diameter 1–3 mm, ring radius scaling 0.1–2 $\times$, section length 0.1–1 $\times$) yield a “wide-and-short” geometric prescription—radial expansion favoured, longitudinal extension penalised; multi-rotor configuration shown as a conceptual render, identified as future work.

Jamieson (personal communication, 2023) derived the multi-rotor mass scaling law $M(1) = 0.577$ (42% saving for equal rotors) and applied it to the RodKite01 geometry. His contribution is cited with gratitude and respect.

Relationship to KTD.jl: The Strathclyde models are steady-state. Chen’s 86.33% efficiency claim is the best TRPT steady-state analysis published. KTD.jl’s 4.2 $\times$ static–dynamic gap is the natural next question: what happens when these configurations are run through full multibody dynamics? Amjad’s multi-rotor concept and “wide-and-short” geometry are precisely what KTD.jl’s DE optimiser produces—the geometric intuition is validated by the optimisation.

4.2 University of Freiburg—Systems Control & Optimization Laboratory

The Freiburg group, led by Prof. Moritz Diehl, has developed three concepts that map directly onto KTD.jl’s architecture.

Leuthold et al. (AWEC 2026) presented a rigidly-convected lifting-line vortex model for AWE optimal control [9]. The core question—“how difficult is it to fly the optimal trajectory found without a wake model, in a real momentum-conserving flow?”—is the academic framing of KTD’s static–dynamic gap. Her 2019 engineering wake induction model for axisymmetric multi-kite systems [7] quantified 18–21% overprediction from neglected induction. Rachel Leuthold is a PhD researcher at Freiburg and co-editor of the AWEC 2026 proceedings.

Diehl, De Schutter & Harzer (AWEC 2026) presented a vertical airborne wind energy farm concept: 99 dual-wing systems stacked at different altitudes, producing 50 MW on 7 km² [10]. The key figure of merit is power density P_D (MW/km²). This is the free-flying analogue of KTD’s multi-rotor TRPT: vertical stacking to increase power per footprint. Dr. Jochem De Schutter is now at TransnetBW GmbH (German transmission system operator), providing an industry grid-integration perspective.

Relationship to KTD.jl: Both groups are solving the same problem—distributing aerodynamic load across vertically-stacked elements—with different mechanical constraints. Leuthold’s rigidly-convected vortex model was designed for embedding a mid-fidelity wake in a fast optimisation loop; KTD’s expansion rotors have a geometrically fixed axisymmetric wake pattern that is a simpler test case than free-flying multi-kite. The 4.2 $\times$ gap is a direct data point for her

research question.

4.3 someAWE Labs—Christof Beaupoil

Christof Beaupoil (someAWE Labs, Alicante) has built the most advanced rotary AWE hardware currently flying [11]. His autogyro pumping-mode system uses a swashplate with active rotation compensation and Kaman-style servo flaps for cyclic pitch control—eliminating pitch links at the rotor hub. This is in cooperation with the University of Freiburg.

Relationship to KTD.jl and CoaxialAutogyroStacking: Beaupoil’s hardware is the closest flying analogue to the autogyro lift subsystem. His swashplate-on-autogyro solution directly addresses the mechanical control problem that CoaxialAutogyroStacking.jl models parametrically. His servo flap approach may simplify the stacked rotor configuration by removing the need for mechanical linkages between units. A visit to Alicante is planned for October 2026.

4.4 Dr. Oliver Tulloch—TRPT Inventor

Oliver Tulloch’s PhD thesis [2] and subsequent Energies paper [1] established the foundational TRPT model: the single-section moment-to-tension ratio ($MTR \approx 0.05$), the spring-disc dynamic formulation, and the $\delta\alpha^*$ collapse criterion. All of KTD.jl’s physics traces back to this work.

Dr. Tulloch now works in offshore wind, optimising legacy wind farm operations in mainland UK. While no longer active in AWE research, his contribution is the foundation on which this work stands, and we cite it accordingly.

Relationship to KTD.jl: KTD.jl extends Tulloch’s single-section MTR to per-section coupling from explicit ring geometry—the moment-to-tension relationship varies along a tapered ring profile. The $\delta\alpha^*$ collapse criterion is implemented dynamically as `collapse_margin =  $\delta\alpha^* - |\Delta\alpha|$` , verified healthy at 42–47° on the left flank.

5 Collaboration Proposals

Windswept’s approach to research collaboration is usually along the lines of: target strategic partners with complementary expertise over funding alone; follow a 10–4–1 pipeline (identify 10, engage deeply with 4, secure 1 meaningful agreement); quantify what each party brings and gains; and frame the technology as a concrete research opportunity. All collaboration proposals below are in-kind exchanges of expertise and tool access—no cash contributions are requested.

5.1 Strathclyde: QBlade cross-validation of KTD aerodynamics

Proposal: A joint paper cross-validating Strathclyde’s QBlade LLFVW aerodynamics against KTD.jl’s ODE solver at matched power scales (12–50 kW). This would be the first multi-fidelity TRPT validation in the literature.

Strathclyde contributes: QBlade aerodynamic analysis of KTD expansion rotor geometries; steady-state TRPT model for comparison. **Windswept contributes:** KTD.jl ODE solver, DE design vectors, multi-rotor campaign data. **Target venue:** *Wind Energy Science* or AWEC 2027.

Possible student project: Co-supervised MSc applying QBlade to KTD.jl expansion rotor configurations. Windswept provides the geometry; Strathclyde provides the aerodynamic analysis.

5.2 Freiburg: Wake induction in multi-rotor TRPT

Proposal: Apply Leuthold’s rigidly-convected vortex wake model to KTD.jl expansion rotor arrays. The fixed-geometry TRPT wake is a simpler test case than free-flying multi-kite—ideal for validating the model on a deterministic wake pattern.

Freiburg contributes: Wake induction model and awebox optimal control framework.

Windswept contributes: TRPT geometry, DE design vectors, multi-rotor campaign data.

Possible student project: Apply Leuthold’s multiple-wake vortex lattice method [8] to KTD.jl’s banked expansion rotor configurations. Co-supervised by Diehl’s group and Read.

5.3 someAWE Labs: Hardware-model cross-validation

Proposal: Cross-validate CoaxialAutogyroStacking.jl’s lift predictions against someAWE’s instrumented flight data. Beaupoil’s swashplate and servo flap experience directly informs the mechanical design of the autogyro lift stack.

someAWE contributes: Flight data, swashplate design experience, hardware reality check.

Windswept contributes: Scaling predictions to 50 kW, autogyro parameter sweep data, stacked configuration analysis.

Planned: Rod Read to visit someAWE Labs in Alicante, October 2026.

5.4 Oliver Tulloch: Attribution and legacy

Proposal: Tulloch is invited to review the KTD.jl model lineage section of any resulting publication, and to accept co-authorship if interested. His spring-disc model is the ancestor of KTD’s multibody dynamics, and this should be documented by its originator.

6 Funding Pathways

Windswept maintains a weekly-updated grant research pipeline (latest: 5 July 2026). Several current opportunities are well-suited to collaborative TRPT research. All figures approximate, in GBP equivalents.

The Horizon Europe D3-03 call (“Wind Energy SET Plan”) is the strongest fit for academic TRPT research. Its Financial Support to Third Parties (FSTP) mechanism is specifically designed for SMEs participating in academic consortia. A September 2026 deadline requires consortium formation within ~4 weeks.

What funding would enable:

- Dedicated researcher time—a postdoc or PhD student focused on QBlade-to-KTD cross-validation

Table 2. Top consortium-ready funding opportunities for TRPT research collaboration.

Grant	Agency	Max award	Deadline	Best fit
EIC Accelerator Open	EIC (EU)	~£2.1M grant	2 Sep / 4 Nov 2026	Commercialisation, TRL 5+ demonstration
Horizon Europe D3-03	EC (EU)	~£340K FSTP Stage 1	15 Sep 2026	Wind energy R&D. FSTP for SMEs in academic consortia.
CETPartnership Call 2026	CETP (EU)	Variable (national pots)	Pre-proposal: 8 Oct 2026	3+ country consortium required. UK eligible via UKRI.
EU Innovation Fund SME	DG CLIMA (EU)	TBC (est. £2–7.5M)	H2 2026	New dedicated SME instrument. Simplified application.

- Hardware validation—instrumented TRPT test rig at a university laboratory
- Travel and workshop costs—regular in-person working sessions (Alicante, Glasgow, Freiburg)
- Open-source development—dedicated Julia developer time for BEM-coupled aerodynamics

The full grant pipeline (17 tracked opportunities) is maintained on the Windswept drive and updated weekly.

7 How to Get Involved

`KiteTurbineDynamics.jl` is open-source (MIT) and actively developed. The quickest way to engage is:

1. **Clone and test:** `git clone https://github.com/rodread/KiteTurbineDynamics.jl`
`&& cd KiteTurbineDynamics.jl && julia -project=. test/runtests.jl`
2. **Read the decisions:** `DECISIONS.md` (2,252 lines) documents every architectural choice from V6 through V10.
3. **Explore the campaigns:** The V10 Tight PCA landscape (Figure 3) and non-dimensional physics atlas (Figure 4) visualise the 14-dimensional design space.
4. **Contact Rod:** `rod@windswept.energy`. Direct email is preferred for initial discussions.

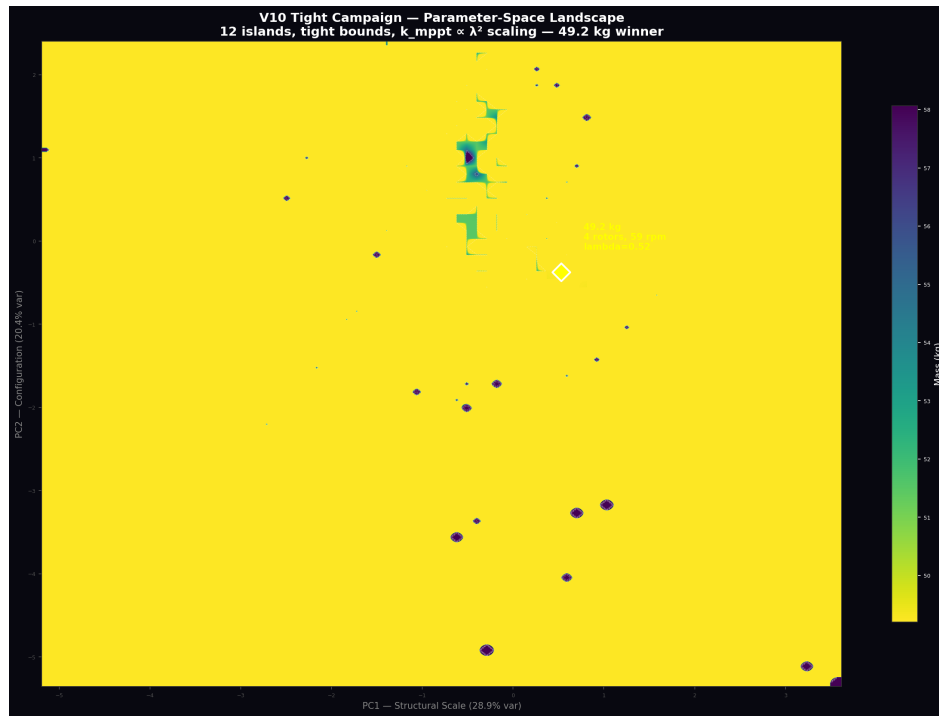


Figure 3. V10 Tight campaign: PCA projection of the 14-dimensional design space. Each point is a design evaluation, coloured by total airborne mass (dark purple = lightest). White dashed contours at 50 and 65 kg. Yellow diamond = campaign winner at 49.2 kg. PC1 (28.9% variance) captures structural scale; PC2 (20.4%) captures configuration choice.

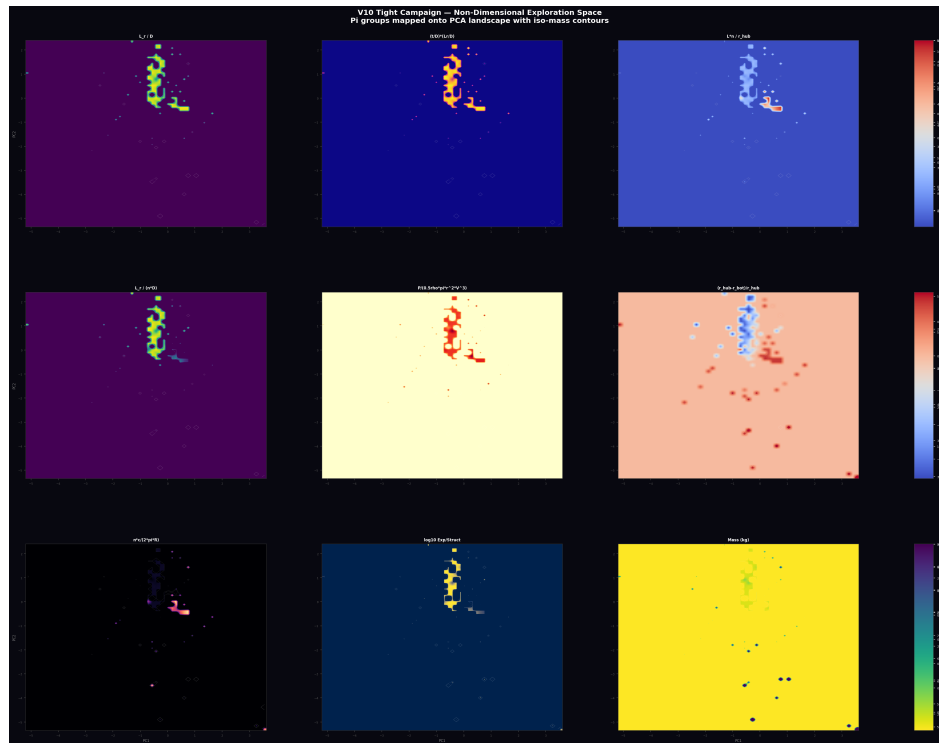


Figure 4. Non-dimensional physics atlas: nine Pi groups colouring the PCA design space, revealing the physical mechanisms behind the 49.2 kg optimum. Panels: beam slenderness, beam efficiency, TRPT aspect ratio, ring packing, power loading, ring taper, rotor solidity, expansion/structural ratio, mass.

8 Conclusion

TRPT kite turbines with multi-rotor expansion represent a material-efficient path to utility-relevant airborne wind energy. The KTD.jl framework has quantified the achievable mass range (49–61 kg at 50 kW), identified the $4.2\times$ static–dynamic gap as the critical modelling challenge, and established left-flank overspeed as the viable operating architecture.

The gaps that remain—mid-fidelity aerodynamics, wake interaction modelling, hardware validation, and dynamic-aware optimisation—are precisely the areas where the groups addressed in this report have world-leading expertise. We invite collaboration on any of the proposals outlined above, and we are prepared to lead or join consortium grant applications to fund dedicated research time.

The simulator is open-source, the test suite is green, and the decisions are documented. The next step is a conversation.

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